Exercise 2

2-1 The circuit shown in Fig.2.1 illustrates a single-channel MOS resistor with a W/L of 2μm/2μm. Using Table.2.1 model parameters calculate the small-signal on resistance of the MOS transistor at various values for V_S and fill in the table below. (Note that the transistor was in linear region, V_B=0, I_{DS}=0)

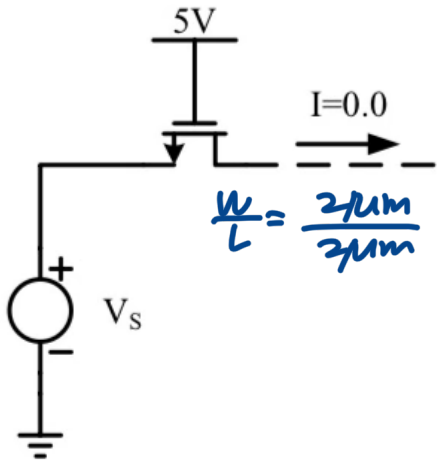


Fig.2. 1

V_S(V)	R(Ω)
0.0	1735.5
1.0	2401.9
2.0	3805.6
3.0	8905.4
4.0	∞
5.0	∞

解. $I=0 \Rightarrow V_{DS}=0$ / $V_{GS}<V_{TH}$,

即处于深三极管区或夹断区

NMOS Model

$V_{T0}=0.7V$, $\gamma=0.45V^{\frac{1}{2}}$

$V_{TH}=V_{TH0}+\gamma(\sqrt{|2\phi_F+V_{SB}|}-\sqrt{2\phi_F})$

$PHI=2\phi_F=0.9V$

$I_D=\mu_n C_{ox}\frac{W}{L}[(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2]$

$K=134\times10^{-6}A/V^2$

$r_{on}=\frac{1}{\partial I_D/\partial V_{DS}}=\frac{1}{K\frac{W}{L}(V_{GS}-V_{TH}-V_{DS})}$

① $V_S=0V$ 时, $V_{TH}=0.7V$, $V_{GS}=5V$

$V_{GS}>V_{TH}$, 深三极管区, $V_{DS}=0V$

$r_{on}=\frac{1}{K\frac{W}{L}(V_{GS}-V_{TH}-V_{DS})}=1735.5\Omega$

② $V_S=1V$ 时, $V_{TH}=0.893V$, $V_{GS}=4V$

$V_{GS}>V_{TH}$, 深三极管区, $V_{DS}=0V$

$r_{on}=\frac{1}{K\frac{W}{L}(V_{GS}-V_{TH}-V_{DS})}=2401.9\Omega$

③ $V_S=2V$ 时，$V_{TH}=1.039V$，$V_{GS}=3V$

$V_{GS}>V_{TH}$，源极跟随管区，$V_{DS}=0V$

$$r_{on}=\frac{1}{K_n'(V_{GS}-V_{TH}-V_{DS})}=3305.6\Omega$$

④ $V_S=3V$ 时，$V_{TH}=1.162V$，$V_{GS}=2V$

$V_{GS}>V_{TH}$，源极跟随管区，$V_{DS}=0V$

$$r_{on}=\frac{1}{K_n'(V_{GS}-V_{TH}-V_{DS})}=3905.4\Omega$$

⑤ $V_S=4V$ 时，$V_{TH}=1.269V$，$V_{GS}=1V$

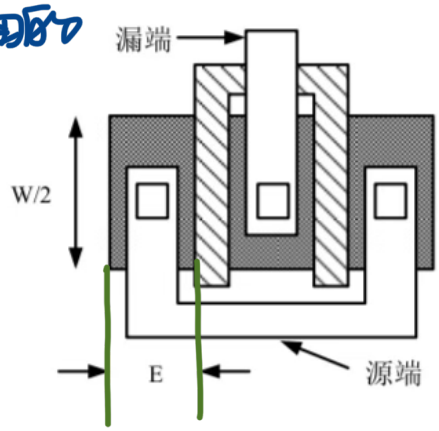
$V_{GS}<V_{TH}$，夹断区，$r_{on}\to\infty$

⑥ $V_S=5V$ 时，$V_{TH}=1.366V$，$V_{GS}=0V$

$V_{GS}<V_{TH}$，夹断区，$r_{on}\to\infty$

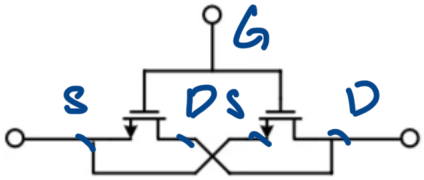
2-2 An NMOS with W=50μm and L=0.5μm operates in the saturated region and its layout is folded shown as Fig. 2.2. Calculate the all capacitances by using the parameters in Table2.2 and $C_{ox}=3.8\times10^{-3}$ F/m, $V_R=0.6$V. Assume that the minimum size (lateral) of S/D region is 1.5μm

假设源漏



E=1.5μm

Fig.2. 2



解：① 栅氧电容 $C_1=\frac{W}{2}L_{eff}C_{ox}=32.3fF$　　$C_{ox}=3.8fF/\mu m^2$

② 耗尽层电容 $C_2=\frac{W}{2}L_{eff}\sqrt{q\varepsilon_{Si}P_{sub}/(4\phi_F)}=7.74\times10^{-7}fF$　　$\frac{W}{L}=\frac{50\mu m}{0.5\mu m}$

③ 交叠电容 $C_3=C_4=\frac{W}{2}C_{ov}$　　$V_R=0.6V$, $2\phi_F=0.9V$

④ 结电容　　$C_{J0}=0.56\times10^{-3}F/m^2$, $m_j=0.45$

$C_{DB}=\frac{W}{2}EC_j+2(\frac{W}{2}+E)C_{jsw}=16.85fF$　　$C_{jsw0}=0.35\times10^{-11}F/m$, $m_{jsw}=0.2$

$C_{SB}=2[\frac{W}{2}EC_j+2(\frac{W}{2}+E)C_{jsw}]$　　$C_j=\frac{C_{J0}}{(1+\frac{V_R}{2\phi_F})^{m_j}}=0.445fF/\mu m^2$

$=WEC_j+(W+2E)C_{jsw}=33.54fF$

$C_{jsw}=\frac{C_{jsw0}}{(1+\frac{V_R}{2\phi_F})^{m_{jsw}}}=3.16\times10^{-3}fF/nm^2$

In saturation region

故 $C_{GS}=2(C_3+\frac{2}{3}C_1)=WC_{ov}+\frac{2}{3}WL_{eff}C_{ox}=63fF$　　$C_{ov}=0.4fF/nm$

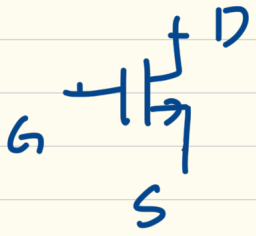
$C_{GD}=2C_4=WC_{ov}=20fF$　　$L_{eff}=L-2L_D=0.34\mu m$

$C_{GB}=2(C_1\|C_2)=1.55\times10^{-6}fF$　　$\varepsilon_{Si}=11.7\times8.85\times10^{-12}F/m$

饱和区，C_{GB}通常被忽略　　$q=1.6\times10^{-19}C$, $P_{sub}=9\times10^{8}m^{-3}$

2-3 There is an N-type current source. I_D is 0.5mA, and the drain-source voltage V_{DS} must more than 0.4V when it works as a current source. If the minimum output resistance is 20 KΩ, determine the length and width of the device by using the parameters in Table.2.2.

解: $\begin{cases} r_o = \frac{1}{\lambda I_D} = 20K\Omega \\ I_D = 0.5mA \end{cases} \Rightarrow \lambda = 0.1$



与Table2.2中λ相等, 故 $L = 0.5\mu m$

$L_{eff} = L - 2L_D = 0.34\mu m$ $\quad K = \mu_n C_{ox} = 134 \times 10^{-6} A/V^2$

current source ⇒ Saturation

$\Rightarrow V_{DSmin} = V_{GS} - V_{TH} = 0.4V$

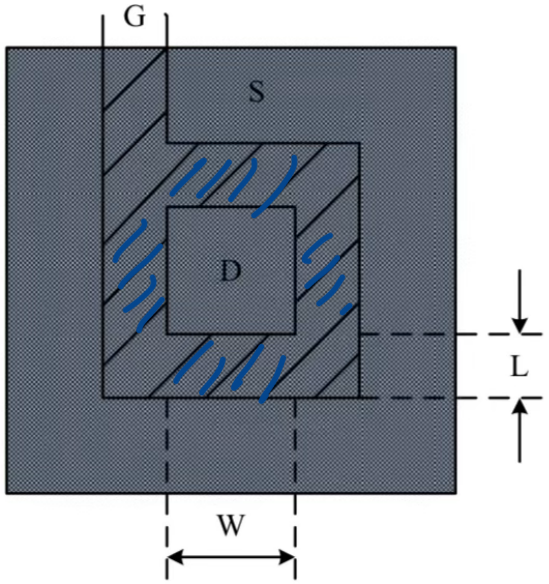
$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L_{eff}} (V_{GS} - V_{TH})^2 = \frac{1}{2} K \frac{W}{L_{eff}} V_{DSmin}^2 = 0.5mA$

$\Rightarrow \frac{W}{L_{eff}} = 46.64 \Rightarrow W = 15.86\mu m$

2-4 A "ring" MOS structure is shown in Fig.2.3. Explain how the device operations and estimate its equivalent aspect ratio. Calculate the drain junction capacitance of the structure. (use Cj and Cjsw)

解: $\frac{W'}{L'} = \frac{4W}{L}$

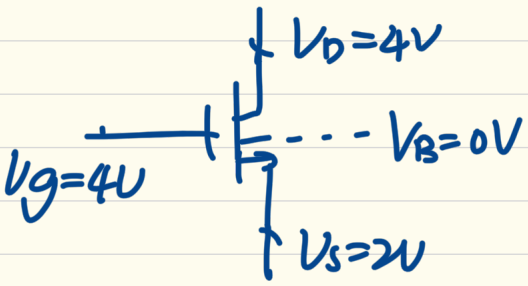
$C_D = W^2 C_j + 4WC_{jsw}$



2-5 Find the small-signal model (g_m, g_{mb}, g_{ds}) for an n-channel transistor with the drain at 4 V, gate at 4 V, source at 2 V, and the bulk at 0 V. Assume the model parameters from Table.2.1, and W/L = 10 μm/1 μm.

解:

$V_{TH}=V_{TH0}+\gamma(\sqrt{|2\phi_F+V_{SB}|}-\sqrt{2\phi_F})=1.039V$



$V_{DS}=2V, V_{GS}=2V$

$V_{DS}>V_{GS}-V_{TH}$, 工作在饱和区

$\frac{W}{L}=\frac{10\mu m}{1\mu m}$

$I_D=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2(1+\lambda V_{DS})$

$g_m=\frac{\partial I_D}{\partial V_{GS}}=\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})(1+\lambda V_{DS})=1.545\times10^{-3}S$

$g_{mb}=\frac{\partial I_D}{\partial V_{BS}}=\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})(1+\lambda V_{DS})(-\frac{\partial V_{TH}}{\partial V_{BS}})$

$\frac{\partial V_{TH}}{\partial V_{BS}}=-\frac{\partial V_{TH}}{\partial V_{SB}}=-\frac{\gamma}{2}(2\phi_F+V_{SB})^{-\frac{1}{2}}$

故 $g_{mb}=g_m\frac{\gamma}{2\sqrt{2\phi_F+V_{SB}}}=1.78\times10^{-4}S$

$K=134\times10^{-6}A/V^2$ $\lambda=0.1$

$\gamma=0.45V^{\frac{1}{2}}$, $2\phi_F=0.9V$

$g_{ds}=\frac{\partial I_D}{\partial V_{DS}}=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2\lambda=6.18\times10^{-5}S$

这题答案认为 $K'=\frac{1}{2}\mu_n C_{ox}=134$

导数答案相较我的 ×2

而且这道题答案对 g_m 也求错了

但同样用 Table2.1，T2.1中写为 $K=\mu_n C_{ox}=134$

$$g_m=\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})(1+\lambda V_{DS})$$
$$=\sqrt{2\mu_n C_{ox}(W/L)I_D}\sqrt{(1+\lambda V_{DS})}$$

Table 2.1中写得K，下面标注 $K'=\frac{1}{2}\mu_n C_{ox}$，PPT里 $K'=\mu_n C_{ox}$

太混乱了，而且Razavi书中好像没有此定义

希望赵老一些